

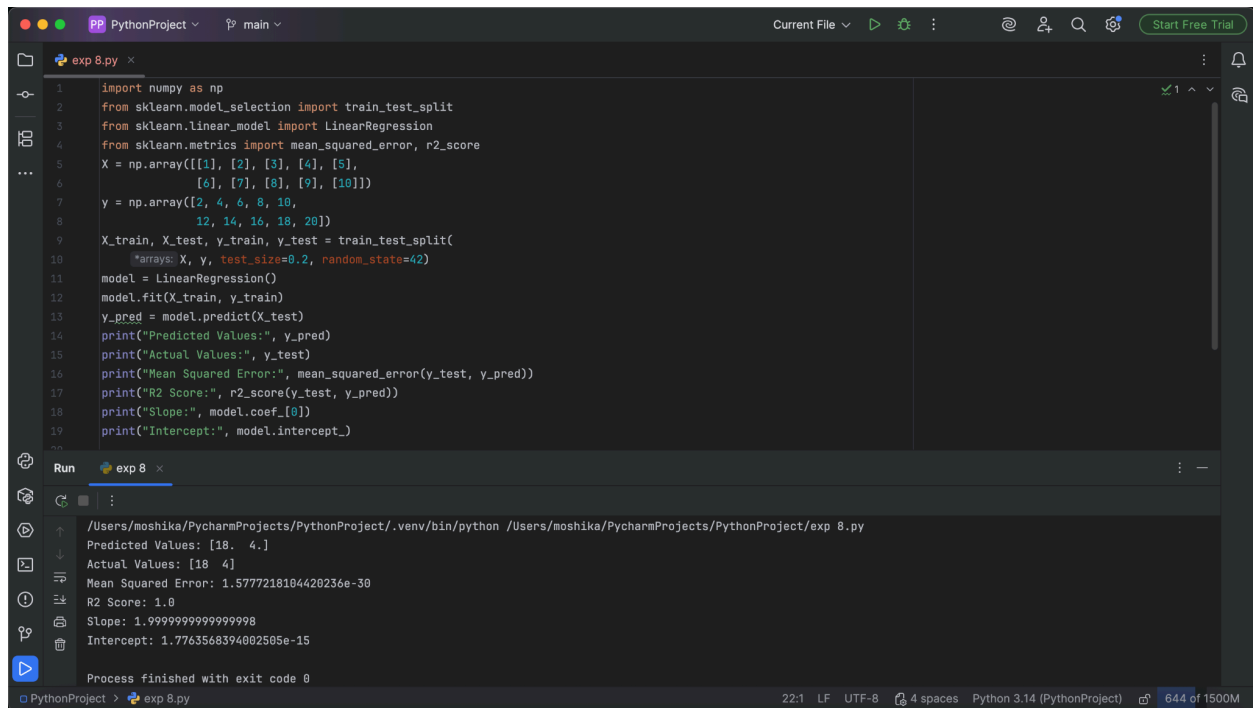
EXPERIMENT - 8

Aim: To implement Linear Regression in Python to predict a continuous output variable.

Procedure:

1. Import the required libraries.
2. Load the dataset.
3. Select the input and output variables.
4. Split the data into training and testing sets.
5. Create a Linear Regression model.
6. Train the model using training data.
7. Predict the output for test data.
8. Calculate the model performance.
9. Display the predicted values and results.

Output:



```
1 import numpy as np
2 from sklearn.model_selection import train_test_split
3 from sklearn.linear_model import LinearRegression
4 from sklearn.metrics import mean_squared_error, r2_score
5 X = np.array([[1], [2], [3], [4], [5],
6              [6], [7], [8], [9], [10]])
7 y = np.array([2, 4, 6, 8, 10,
8              12, 14, 16, 18, 20])
9 X_train, X_test, y_train, y_test = train_test_split(
10     X, y, test_size=0.2, random_state=42)
11 model = LinearRegression()
12 model.fit(X_train, y_train)
13 y_pred = model.predict(X_test)
14 print("Predicted Values:", y_pred)
15 print("Actual Values:", y_test)
16 print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
17 print("R2 Score:", r2_score(y_test, y_pred))
18 print("Slope:", model.coef_[0])
19 print("Intercept:", model.intercept_)
20
```

Run exp 8

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 8.py
Predicted Values: [18.  4.]
Actual Values: [18  4]
Mean Squared Error: 1.5777218104420236e-30
R2 Score: 1.0
Slope: 1.9999999999999998
Intercept: 1.7763568394002505e-15
Process finished with exit code 0
```

Result: The Linear Regression model successfully predicted the continuous target values.